Proposal Defense

Socratic AI, Agentic AI for Fostering Critical Thinking Education

A multi-agent tutor that asks instead of answers, and measures how well the learner reasons

Presented by Muhammad Eusoff Amir bin Hazri (22010491)

Supervised by Dr Ho Tatt Wei

Department of Electrical and Electronics Engineering

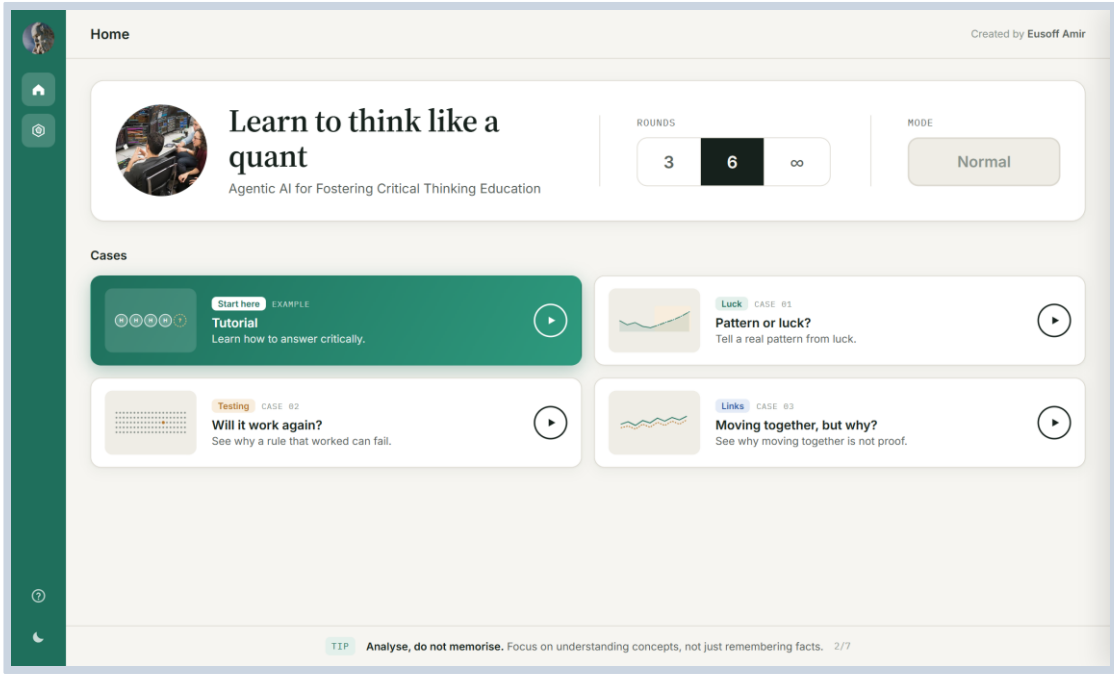
Course EFB4042 Final Year Project I

Project code 26S005

Date 21 October 2026

OUTLINE OF PRESENTATION

1. Introduction
2. Problem Statement, Objectives, Scope of Study
3. Literature Review
4. Proposed Methodology
5. Gantt Chart & Milestones
6. Preliminary Results
7. Conclusion



The working prototype, live online

INTRODUCTION

Students can ask Gen AI for a finished answer and copy it in seconds, so they skip the thinking.

62% of students

use generative AI at least weekly [12]

319 workers studied

says more trust in AI meant less critical thinking [10]

4 costs of "outsourced reasoning"

- fewer ideas,
- less effort,
- weaker memory,
- less reflection [11]

The idea of this project

Turn the same AI around. Instead of an answer machine, build a tutor that asks Socratic questions in a set order, so the student builds the understanding, and then measure the quality of that reasoning.

PROBLEM STATEMENT

Gen AI can now do a lot of the analysis and coding work. So, engineering students need strong critical thinking, and use AI to help them think, not to think for them [10]. Some Socratic AI tutors already avoid giving answers, but they still have three gaps.

1

No ladder

The questions only lead to an answer, and learners never check if it is the right question.

2

No measurement

They do not measure the learner's critical thinking against any framework.

3

No adaptation

They do not adapt to what each learner is missing.

A tester of two Gemini Gem tutors wrote *"I've explained this before"*

So, there is a need for an agentic AI tool that helps engineering undergraduates build their critical thinking.

OBJECTIVES

1. **Teach** students new to a topic build their own reasoning. The tutor guides them up Paul's six levels of questions and does not give the answers.

2. **Assess** each student's critical thinking from the chat. It uses the Paul-Elder framework, and its marks agree with human markers.

3. **Adapt** to match each student's gaps. Students can see what they have mastered and what to work on next.

These match the three objectives and deliverables in the official project listing (26S005). The topic (quant) was set by the supervisor.

SCOPE OF STUDY

1

Any topic, quant checked

Quant is the checked topic, set by the supervisor.
Telling a real pattern in data from luck, for beginners, in 3 short cases.
Any other topic is drafted by a Topic Builder agent.

2

Multi-agent tutor

AI agents (Assessor, Planner, Tutor, Checker, Coach, Reasoning, Explainer, Topic Builder) guarded by plain-code rules, built with LangGraph. Free Gemini for now, Claude AI supported.

3

Framework assessment

6 Socratic levels, 9 standards, reasoning type and knowledge type, with a mastery dashboard.

Target learners are university students new to quant, and the first testers are Dr Ho's students.

In FYP2, I check the agents against human markers, add real quant reading with RAG [14] (Wikipedia stands in for now), and run a full class study after ethics approval.

LITERATURE REVIEW

Agentic AI Building Blocks

Technique	Year	Core idea	Use in this project	Ref
Prompting	2025	A prompt is an instruction plus context; output depends on how the context is built	Every agent gets a structured JSON prompt	[15]
Chain of Thought	2022	Break a task into intermediate reasoning steps	The Planner reasons before choosing the next move	[16]
ReAct	2023	Reason, act, observe, then repeat	The Tutor runs a simulation, reads the result, then replies	[17]
RAG	2023	Ground answers in retrieved knowledge (naive, advanced, modular)	Case notes now; RAG with a knowledge graph planned for FYP2	[14], [18]
Model Context Protocol	2024	A standard way to give an LLM tools, resources and prompts	Later, to query a knowledge graph and learner records	[19]
Multi-agent systems	2024	Specialised agents collaborate (MetaGPT, AgileCoder)	Assessor, Planner, Tutor and Checker roles	[20], [21]

Based on the supervisor's course slides [15], [22] and the DeepLearning.AI short courses he recommended [23].

LITERATURE REVIEW

Frameworks for Critical Thinking

Framework	Year	Core idea	Role in this project	Limitation	Ref
Paul's 6 Socratic question levels	2006	Clarify, assumptions, evidence, viewpoints, implications, question the question	The ladder of what the Tutor asks next	Says what to ask, not how good the answer is	[1]
Paul-Elder 9 intellectual standards	2019	Clarity, accuracy, precision, relevance, depth, breadth, logic, significance, fairness	The Assessor marks each as 1 or 0	Does not say which type of reasoning was used	[1]
Paul-Elder 8 elements of thought	2019	Purpose, question, information, inference, concepts, assumptions, implications, point of view	Prerequisites before reasoning (planned)	Links to reasoning types still open	[1]
Revised Bloom's taxonomy	2001	Knowledge types, such as factual, conceptual, procedural and metacognitive	Types of knowledge used (planned)	No causal or relational type	[2]
SOLO taxonomy	1982	Prestructural to extended abstract outcomes	How complex an answer is, by how many knowledge types are joined	Needs adapting to dialogue	[3]

LITERATURE REVIEW

Socratic Questioning and Socratic AI

Study	Year	Setting	Main finding	Limitation	Ref
Sahamid	2016	Classroom action research	Socratic questioning built critical thinking	Small, one class	[4]
Makhene	2019	Nursing education	Socratic inquiry helps critical thinking	Needs skilled teachers' time	[5]
Ho, Chen and Li	2023	Healthcare students	Socratic method improved thinking skills	Human tutor, does not scale	[6]
Degen and Asanov	2025	65 pre-service teachers, Socratic AI vs plain chatbot	More support for critical, reflective thinking; argues for multi-agent systems	Self-reported, not a direct measure	[7]
Fan	2025	Socratic scaffolding through AI	Structured scaffolding deepened inquiry	No link to reasoning constructs	[8]
Lee, Ang and Lai	2025	Bloom labels on chatbot dialogues	Dialogue logs reveal thinking processes	Descriptive only	[9]
Lee et al. (CHI)	2025	319 knowledge workers	More AI trust, less critical thinking	Survey, self-reported	[10]

LITERATURE REVIEW

Existing Socratic AI Tutors

Tool	Refuses answers	Question ladder	Framework assessment	Goal, progress, end	Reasoning and knowledge types
Online Gemini Gem "Socratic Tutor" (Test #1)	Yes	No	No (informal points)	No	No
Project prompt v1 (Test #2)	Yes	Partly	Partly (hidden 1 to 3 score)	No	No
Khanmigo, custom GPTs *	Yes	Not shown	Not shown	Partly	No
Socratic AI (this project)	**Yes**	**Yes, 6 levels**	**Yes, 9 standards**	**Yes**	**Yes**

** From public descriptions, not tested in this project.*

No tested tool combines a question ladder, framework assessment and a clear session, and none shows which type of reasoning a learner used.

LITERATURE REVIEW

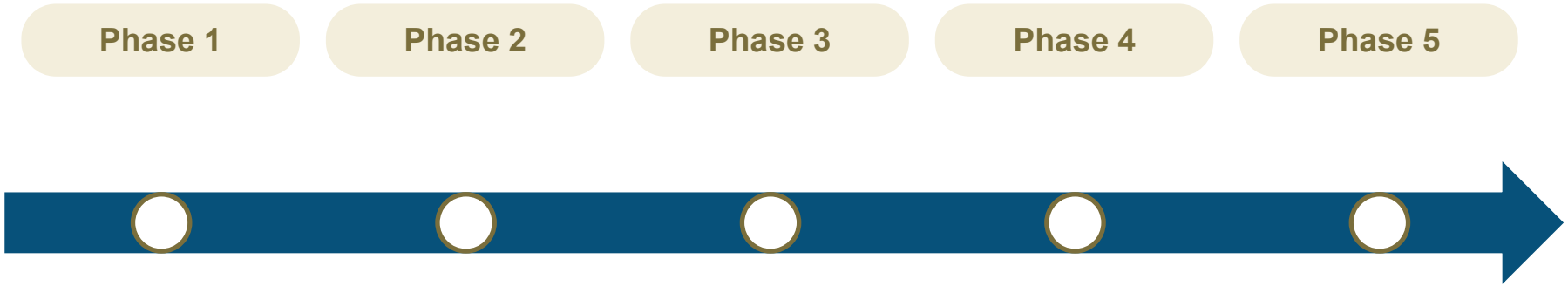
RESEARCH GAP

- Using AI can make students think less, and most AI tutors just give answers.
- Socratic AI tutors help, but their questions do not follow a clear step-by-step order.
- Most studies do not check reasoning against a set framework. They mostly rely on what students say about themselves.
- No tool shows students which type of reasoning or knowledge they used.

JUSTIFICATION

- A multi-agent tutor where the Tutor follows Paul's six levels, so the questions go up in a set order.
- A separate Assessor marks each answer on the nine standards and gives evidence. We can then compare its marks with human markers.
- Simple code rules check every AI decision, so each step is fair and testable.
- Next, new agents will also mark the type of reasoning and knowledge.

PROPOSED METHODOLOGY



Early testing

Two Gemini Gem tutors tested; problems written down

Requirements

9 design requirements R1 to R9 from the tests

Build

AI agents, rules, cases, simulations, web app

Test and fix

Real sessions read turn by turn; problems fixed

Evaluate

Requirements, Assessor vs humans, learners

R1 Clear mission	R2 Fixed rounds, progress	R3 Remember, never re-ask
R4 Feedback and teaching	R5 Notice frustration	R6 Vary formats
R7 Separate assessor	R8 Learner stats	R9 Stable topic

Software

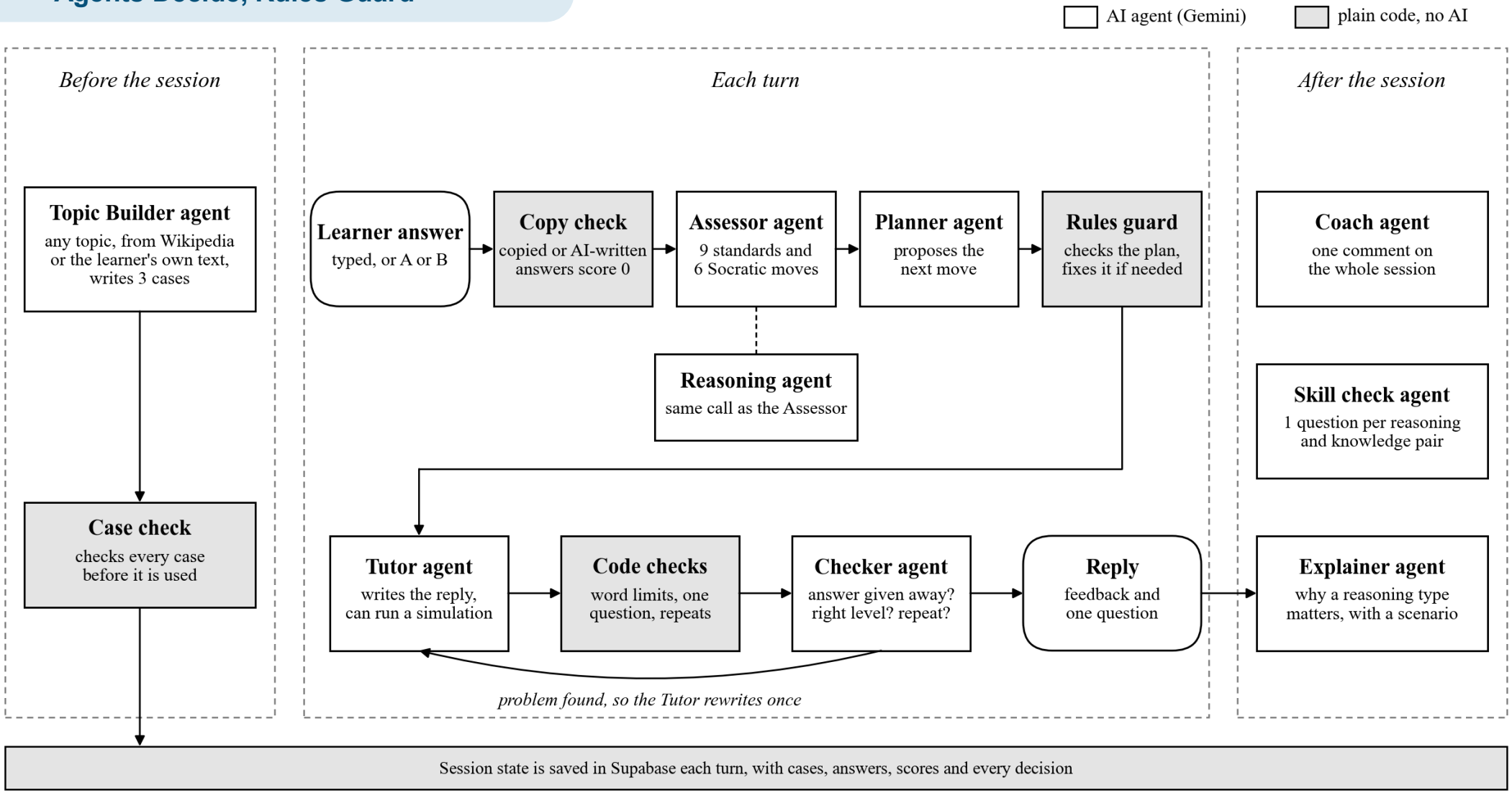
- Python 3.13, FastAPI, LangGraph
- Claude AI (as listed) is supported; Gemini 3.5 Flash Lite (free tier) is used for now
- HTML and JavaScript front end
- Supabase database, Render hosting (free)
- 196 automatic tests with pytest

Hardware

- A normal laptop; the AI runs in the cloud, so no GPU is needed

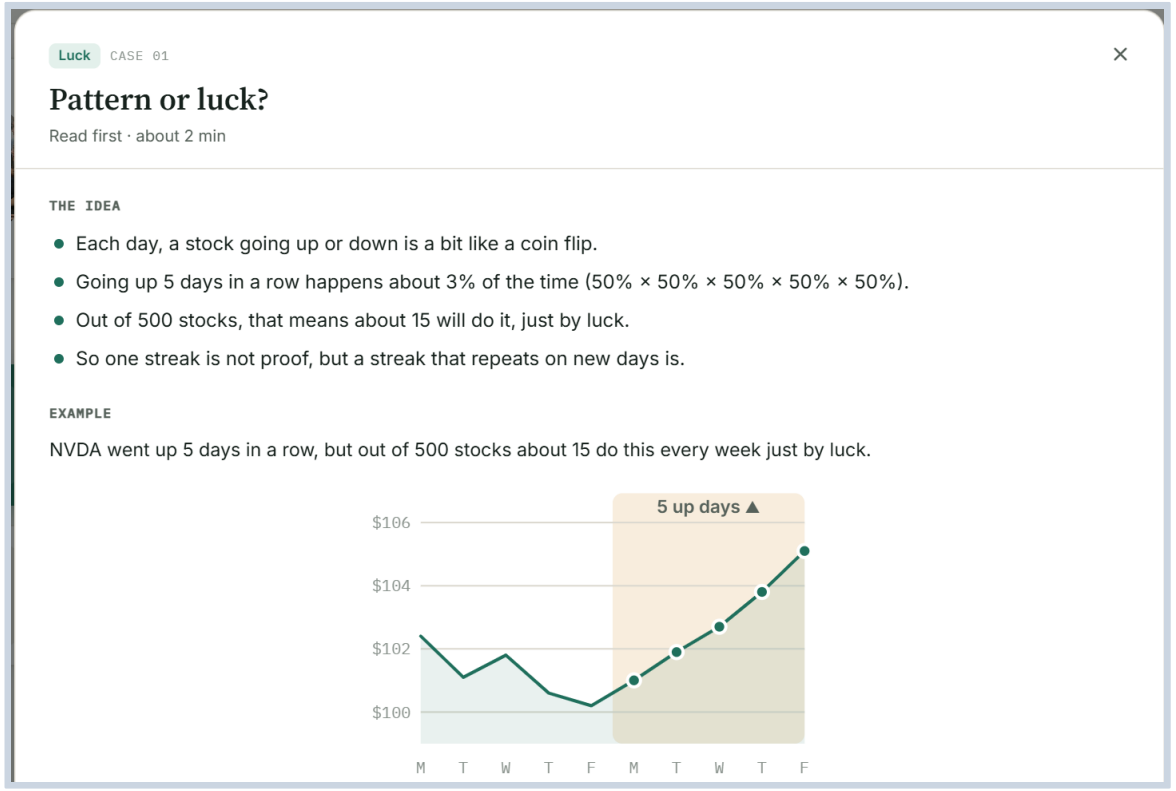
PROPOSED METHODOLOGY

Agents Decide, Rules Guard

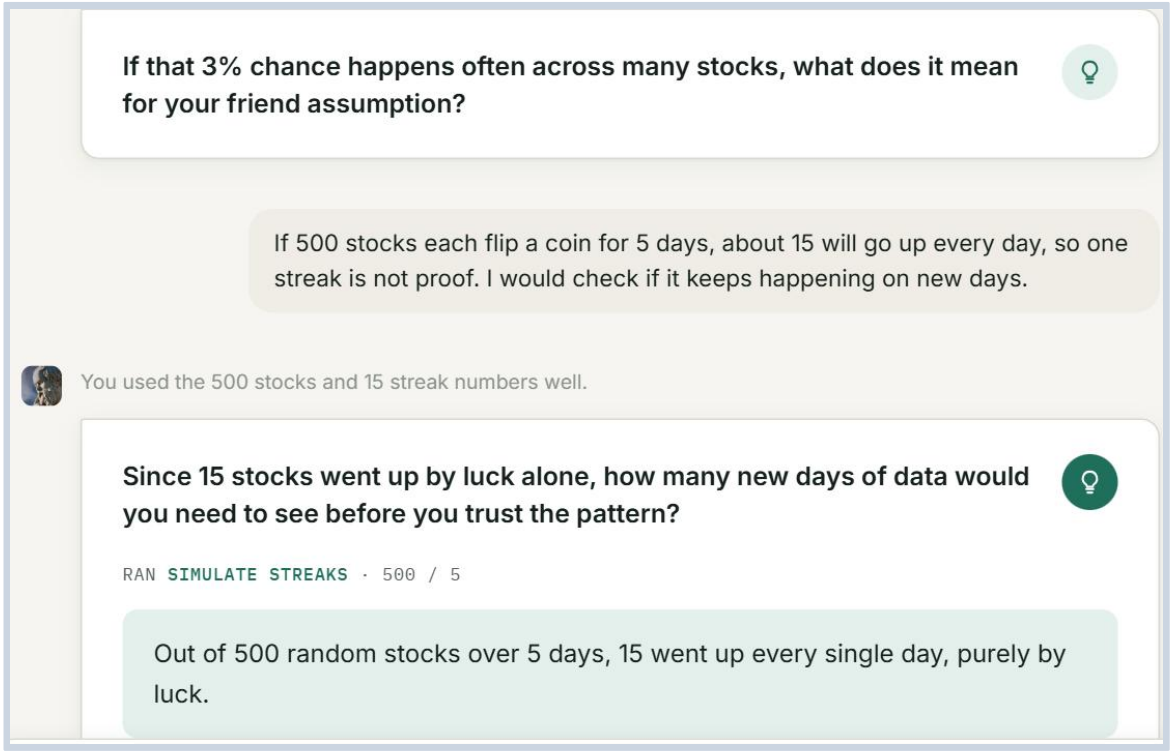


PROPOSED METHODOLOGY

Session Design (R1 to R9)



1. A short reading and a mission before each case

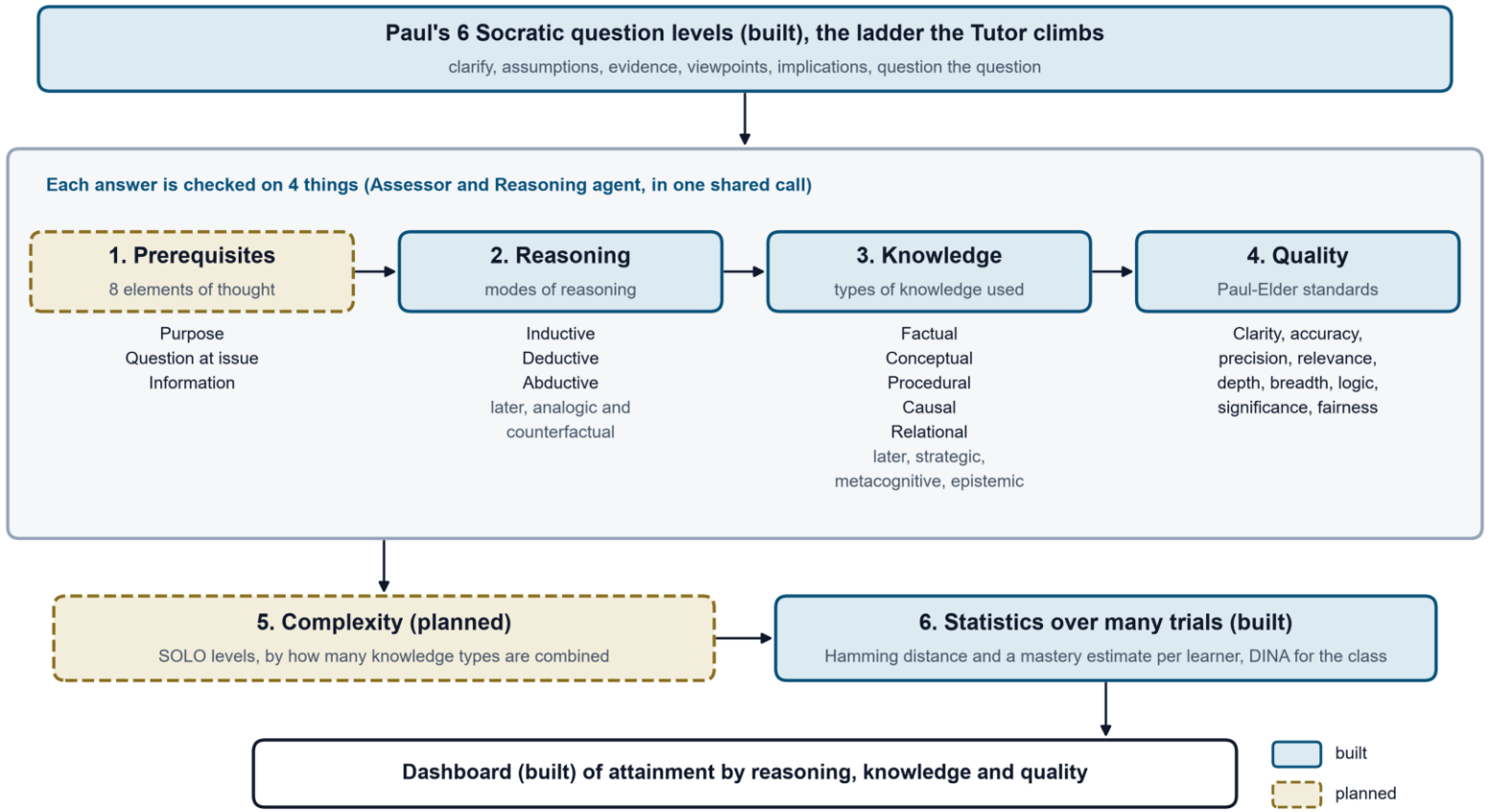


2. Feedback, one question, a hidden hint, and a simulation result

3, 6 or no-limit rounds | **Round and level shown** | **4 question formats** | **"I'm not sure" button** | **Status panel** | **Tutorial**

PROPOSED METHODOLOGY

Reasoning and Knowledge Layer (Goal 1, built)



For example, "All the regional banks fell, not just CIMB" is inductive reasoning on relational knowledge.

- **Paul's 6 levels ask the questions.** The Tutor asks questions using Paul's 6 levels, and the student replies.
- **1 to 4 check one answer.** For each answer, we ask if they had the basics (1), how they reasoned (2), what knowledge they used (3) and how good it was (4). Each one gets a 1 or 0.
- **Complexity (planned) uses the result from Knowledge**. If the answer used only one knowledge type, it is simple. If it combined three, it is more complex.
- **Statistics collects everything**. Each answer gives a row of 1s and 0s. Statistics counts the rows to find the skills shown often, and the Hamming distance [13] counts the skills left.

PROPOSED METHODOLOGY

Evaluation Plan

1 Requirement check

Objective 1

Run the same scenarios on the baseline Gem tutors and on Socratic AI, and mark R1 to R9 as met or not.

2 Assessor vs human markers

Objective 2

Mark 61 test answers by hand with a marking guide. Compare using percent agreement and Cohen's kappa per standard (parent grant target 0.8).

3 With and without agents

All objectives

Switch the Planner, simulations or Checker off, and compare the same sessions.

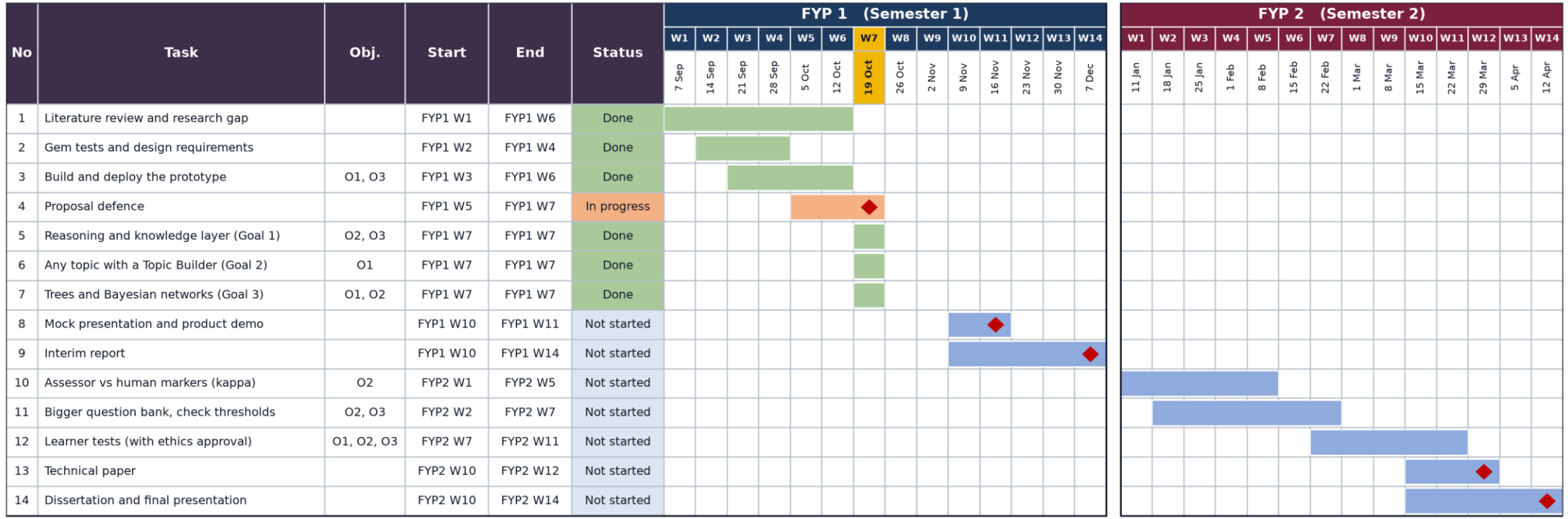
4 Learner experience

Objective 3

A short questionnaire with Dr Ho's students, after ethics approval.

GANTT CHART & MILESTONES

No	Task	Obj.	Start	End	Status
1	Literature review and research gap		FYP1 W1	FYP1 W6	Done
2	Gem tests and design requirements		FYP1 W2	FYP1 W4	Done
3	Build and deploy the prototype	O1, O3	FYP1 W3	FYP1 W6	Done
4	Proposal defence		FYP1 W5	FYP1 W7	In progress
5	Reasoning and knowledge layer (Goal 1)	O2, O3	FYP1 W7	FYP1 W7	Done
6	Any topic with a Topic Builder (Goal 2)	O1	FYP1 W7	FYP1 W7	Done
7	Trees and Bayesian networks (Goal 3)	O1, O2	FYP1 W7	FYP1 W7	Done
8	Mock presentation and product demo		FYP1 W10	FYP1 W11	Not started
9	Interim report		FYP1 W10	FYP1 W14	Not started
10	Assessor vs human markers (kappa)	O2	FYP2 W1	FYP2 W5	Not started
11	Bigger question bank, check thresholds	O2, O3	FYP2 W2	FYP2 W7	Not started
12	Learner tests (with ethics approval)	O1, O2, O3	FYP2 W7	FYP2 W11	Not started
13	Technical paper		FYP2 W10	FYP2 W12	Not started
14	Dissertation and final presentation		FYP2 W10	FYP2 W14	Not started

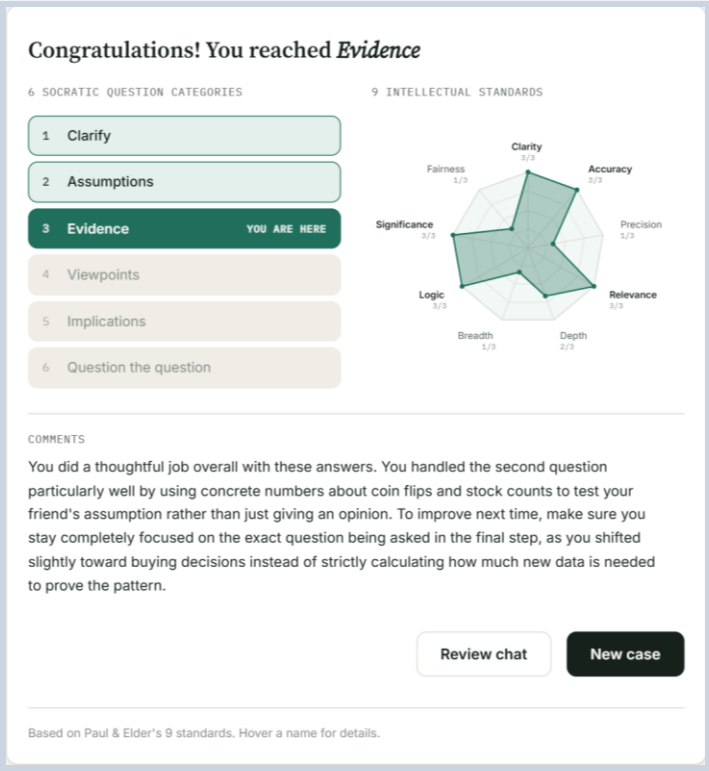


O1 to O3 are the three objectives (teach, assess, adapt)

- **21 Oct 2026 (FYP1 W7)** — Proposal defence
- **16 Nov 2026 (FYP1 W11)** — Product demo, mock presentation
- **7 Dec 2026 (FYP1 W14)** — Interim report
- **29 Mar 2027 (FYP2 W12)** — Technical paper
- **12 Apr 2027 (FYP2 W14)** — Dissertation, final presentation

PRELIMINARY RESULTS

A Real 3-Round Session (Gemini)



The end profile shows the level reached, the 9 standards chart and the Coach comment

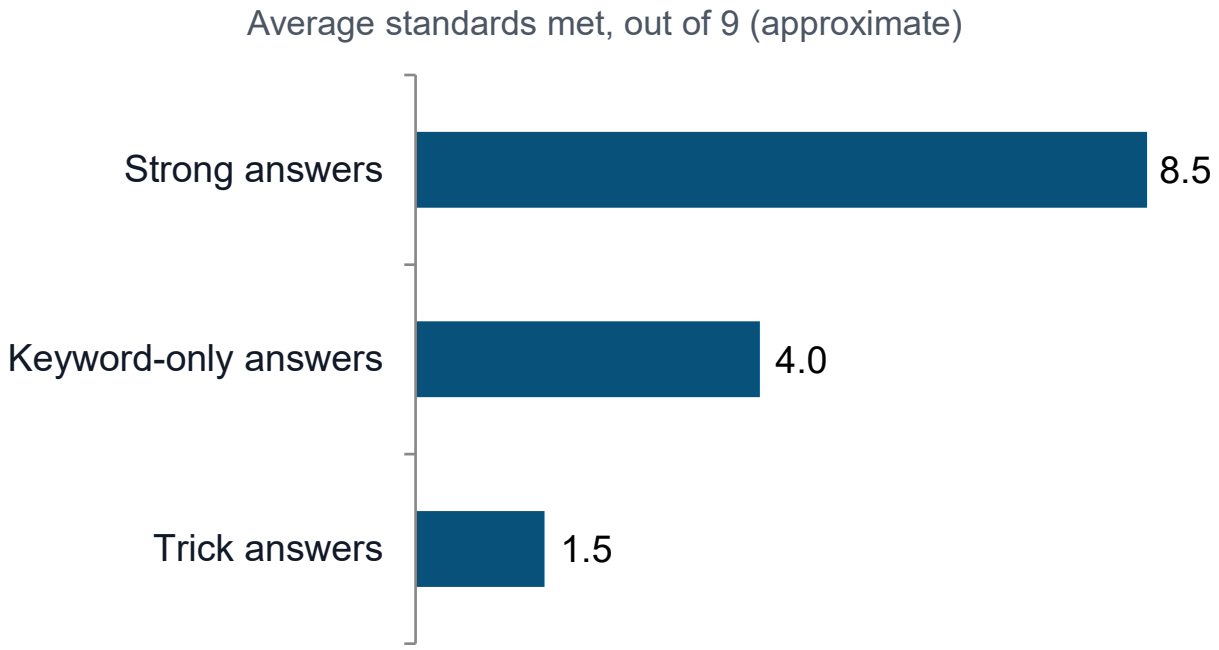
0 times the answer was given away, even when asked to ignore the rules

5 of 5 problems found in testing were fixed

~24 AI calls per session, inside the free tier

PRELIMINARY RESULTS

Assessor on a 61-Answer Test Set



37 real answers from test sessions and 24 constructed answers. Not yet compared with human markers, so this is not a validation yet.

Problems found and fixed

- The welcome said there was no round limit, so code now writes the limit
- The same question came back in new words, so the Checker now compares questions
- The Planner's simulation was ignored, so the rules now run it
- Levels rose too fast, so a level up now needs 2 good answers in a row or 1 strong answer
- The Assessor was too generous, so each standard now has strict rules, and standards met fell from 24 to 16 on 12 real answers

Next, keyword-only answers still score about 4 of 9, so I will make the Assessor prompt stricter.

CONCLUSIONS

Other Socratic AI tutors have no **clear question order**, no **measurement** and no **adaptation**.

Socratic AI fixes all three by asking questions up Paul's six levels, marks each answer on the nine standards and adapts to each student.

It now also shows the **type of reasoning** and **knowledge used** on any topic. Next, its marks will be checked against human markers.

Take-home message

Socratic AI is an education tool, not an answer machine. It is built with several prompted AI agents that work together, and simple code rules keep them in check.

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